

ColorRef: Interactive Reference Games for Probing the Behavioral Geometry of Color Grounding

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Abstract

We introduce COLORREF, an interactive reference-game framework for evaluating language-to-color grounding. Built from 518,830 English color descriptions paired with sRGB and CIELAB targets, COLORREF asks models to predict a color, receive natural-language feedback, and revise their guess over multiple rounds. By comparing oracle and LLM teachers, the framework separates one-shot prediction, feedback following, feedback bandwidth, and feedback generation. Controlled oracle feedback substantially improves accuracy and behavioral geometry: Qwen3-14B improves from 40.58 to 22.32 mean ΔE_{76} , while distance-geometry alignment rises from 0.461 to 0.796. Sequential interaction outperforms same-budget single-turn feedback, and abstract descriptions benefit most. LLM-teacher experiments show that models can follow valid perceptual feedback better than LLMs can generate it, exposing feedback generation as a distinct bottleneck.

1 Introduction

Color descriptions are a compact testbed for grounded language. Some names map directly to perceptual regions, as in *dark green* or *pale blue*; others refer through objects and materials, as in *ocean blue* or *mustard yellow*; and many are associative or idiosyncratic, as in *abandoned hope*, *woman of the dunes*, or *villainy purple*. These expressions are linguistic, but their referents lie in a continuous perceptual space. Color therefore lets us ask whether models can do more than associate words with labels: can they use language to navigate perceptual structure?

Grounded communication is rarely a single prediction. People often negotiate color reference through corrections such as *lighter*, *warmer*, *more muted*, or *less blue*. Such feedback is linguistically simple but geometrically meaningful: it specifies local directions in a shared perceptual space.

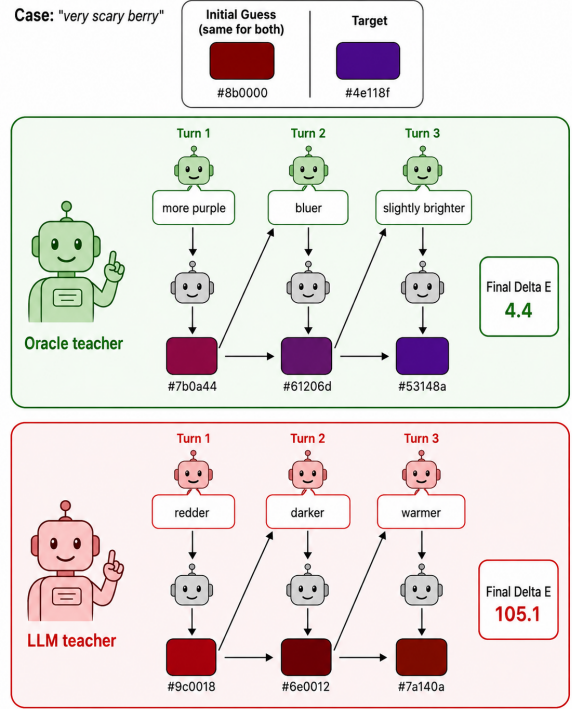


Figure 1: Paired trajectory for “very scary berry”. Both conditions start from the same initial guess. Oracle feedback gives geometrically valid corrections, steering the guess toward the purple target ($\Delta E_{76} = 4.4$), whereas LLM feedback is fluent but misdirected, leaving the guess in the red region ($\Delta E_{76} = 105.1$). The example illustrates the gap between following valid feedback and generating valid feedback.

A model that guesses poorly at first may still be grounded if it revises in the right direction; conversely, a model may produce fluent feedback that points another agent away from the target. One-shot evaluation cannot distinguish these cases.

Most language-to-color evaluations are one-shot: given a description, a model predicts a color, retrieves a candidate, or chooses among alternatives (McMahan and Stone, 2015; Monroe et al., 2017). Recent work also compares language-model representations with perceptual color geometry (Abdou

Regime	Description	Swatch
Explicit	faded old rose black as coal rich purple paint	
Prototype	the green of limes sweet orange sunset montmorency cherry	
Compound	gravel grey cryogenic electric blue fresh kiss red	
Abstract	woman of the dunes ketchup and mayo mix under the seafoam blue	

Table 1: Sample color descriptions from the COLORREF corpus, grouped by semantic regime. Swatches show crowd-aggregated ground-truth sRGB targets.

et al., 2021; Loyola et al., 2023). These approaches test whether initial predictions or representations align with color space. We ask a complementary behavioral question: can models use natural-language feedback to move through color space, and can they generate feedback that helps another model do so?

We introduce COLORREF, an interactive reference-game framework for probing color grounding in language models. In each game, a model receives a natural-language color description and predicts an sRGB hex code. The prediction is converted to CIELAB space and compared with a hidden target. A teacher then provides natural-language feedback, and the model revises its guess for up to three rounds. The resulting trajectory, illustrated in Figure 1, lets us evaluate final accuracy, movement direction, feedback-constraint satisfaction, and alignment between model behavior and perceptual color geometry.

A key design choice is that the teacher is an experimental condition rather than an uncontrolled conversational partner. Deterministic oracle teachers emit controlled corrections along perceptual axes such as lightness, red–green, yellow–blue, and saturation, isolating feedback following. We vary bandwidth—one, two, or three constraints per turn—and compare canonical axis-style messages with natural template paraphrases. We also evaluate LLM teachers, which must generate feedback themselves. This decomposes interactive grounding into initial prediction, feedback following, bandwidth, wording, and feedback generation.

We build COLORREF from 518,830 English color descriptions paired with sRGB colors, RGB/HSV/CIELAB coordinates, and semantic-

regime labels. Table 1 shows examples from four regimes: *explicit grounded*, *prototype mediated*, *compound associative*, and *abstract idiosyncratic*. These regimes test whether interaction helps differently when a description directly names a color versus indirectly evokes one.

We report color error as ΔE_{76} , the Euclidean distance between predicted and target colors in CIELAB space (Colorimetry, 1986). Because our task asks models to infer a crowd-sourced target from a short description, we treat ΔE_{76} as a perceptual-space distance rather than a strict just-noticeable-difference measure: lower is better, but even improved predictions may remain visually distinct from the exact target.

Our experiments show that interaction improves color grounding in ways that one-shot prediction cannot capture. Controlled oracle feedback roughly halves mean error on the 12,000-example scale-up set, with Qwen3-14B improving from 40.58 to 22.32 ΔE_{76} under three-constraint feedback. Ablations show that feedback bandwidth matters, but that interaction is not just a larger information payload: at matched total constraint budgets, sequential feedback outperforms delivering the same constraints in one turn. The benefits are strongest for abstract descriptions, which are hardest in one shot but substantially recoverable through feedback.

The same experiments reveal a teacher-side bottleneck. When feedback is geometrically valid, the guesser can often follow it, improving both final accuracy and global alignment with LAB geometry. LLM teachers, however, frequently produce fluent feedback that points in the wrong perceptual direction. Thus, COLORREF separates two capabilities that are often conflated in interactive settings: using feedback to correct a trajectory, and generating feedback that is itself geometrically correct.

Together, these results show that interactive grounding is a stronger test than one-shot prediction: models must not only name a point in color space, but use language to correct their trajectory through it. COLORREF makes this behavior measurable, revealing a clear gap between following perceptual feedback and generating it.

2 Related Work

Color language, grounding, and language models. Color has long served as a testbed for links between language, perception, and categorization. Classic work identifies cross-linguistic regularities

in basic color terms (Berlin and Kay, 1991), while later accounts model color naming through perceptual geometry and communicative efficiency (Regier et al., 2007; Zaslavsky et al., 2018). In NLP, grounded color semantics has been studied through probabilistic and neural models that map linguistic expressions to perceptual observations (McMahan and Stone, 2015; Monroe et al., 2017). Recent work asks whether language models encode perceptual color structure without direct grounding: Abdou et al. (2021) compare monolexemic color-term representations with CIELAB geometry, and Loyola et al. (2023) show that alignment degrades with subjectivity and abstractness. COLORREF extends this line from static representations and one-shot predictions to feedback-driven behavioral trajectories.

Comparatives, reference games, and pragmatics. Our protocol builds on comparative color language and reference games. Winn and Muresan (2018) model terms such as *lighter* and *darker* as directions in color space, while Monroe et al. (2017) study pragmatic color reference as target identification among contextual alternatives. More broadly, reference games and Rational Speech Act models treat communication as speaker–listener coordination (Lewis, 2008; Frank and Goodman, 2012; Goodman and Frank, 2016), and grounded clarification work distinguishes perceptual from collaborative grounding (Benotti and Blackburn, 2021). COLORREF adapts these ideas to continuous correction: the relevant context is the model’s current guess, not a fixed distractor set.

Feedback, revision, and LLM teachers. Interactive grounding work argues that models should be evaluated not only by task success, but by how common ground is built over time (Imai et al., 2025). Feedback is also central to instruction tuning, RLHF, AI feedback, critique, and iterative revision (Ouyang et al., 2022; Bai et al., 2022; Saunders et al., 2022; Madaan et al., 2023; Shinn et al., 2023). In grounded settings, however, feedback quality is itself a source of error: a teacher may produce fluent guidance that is not perceptually correct. COLORREF therefore separates the ability to follow feedback from the ability to generate it, using oracle and LLM teacher conditions.

Regime	Count	%
Abstract Idiosyncratic	238,416	46.0
Prototype Mediated	137,420	26.5
Compound Associative	94,246	18.2
Explicit Grounded	48,748	9.4

Table 2: Distribution of semantic regimes in the full COLORREF corpus.

3 The COLORREF Framework

COLORREF evaluates whether language models can ground short color descriptions in continuous perceptual space through interaction. Each example contains a natural-language color description, a target sRGB color, and corresponding perceptual coordinates. A model first predicts a color from the description alone, then revises its prediction after receiving feedback. The full sequence of guesses supports evaluation of both final accuracy and trajectory-level behavior.

3.1 Dataset

We build COLORREF from English color descriptions paired with sRGB hex codes derived from the ColorNames website¹. After preprocessing and filtering, the corpus contains 518,830 examples. Each row contains the description, target hex code, RGB, HSV, and CIELAB coordinates, parser outputs, and semantic-regime annotations. We use CIELAB as the main evaluation space because distances in LAB are more perceptually meaningful than raw RGB distances.

Each example is written as (x_i, h_i, c_i, r_i) , where x_i is the color description, h_i is the target sRGB hex code, $c_i \in \mathbb{R}^3$ is the corresponding LAB coordinate, and r_i is a semantic-regime label. The guesser observes only x_i at the initial turn; h_i and c_i are hidden from the guesser and used only by teachers and evaluators. Details on the preprocessing steps in Appendix C.1, C.3.

3.2 Semantic Regimes

Color names vary in how directly they constrain the target color. We assign each description to one of four mutually exclusive semantic regimes using a lightweight parser with component scores for explicitness, prototype anchoring, rarity, and abstraction (See details in Appendix C.2). These labels are coarse analysis categories, not exhaustive semantic parses.

¹colornames.org, published under CC0 1.0 license

We use four regimes: **explicit grounded** descriptions directly name a color or modifier, as in *dark green*; **prototype mediated** descriptions refer through objects or materials, as in *ocean blue*; **compound associative** descriptions combine a partial color anchor with additional content, as in *corporate blues*; and **abstract idiosyncratic** descriptions are affective or metaphorical, as in *abandoned hope*. These regimes separate description validity from directness: all examples are plausible color descriptions, but differ in how explicitly they identify a region of color space.

3.3 Evaluation Subsets

We construct three evaluation subsets from the full corpus. `main_12000` is the primary scale-up subset, with 12,000 examples balanced across regimes. `main_4000` contains 4,000 examples, balanced across regimes and hue, lightness, and saturation bins. `debug_4000` contains 4,000 examples balanced across regimes and is used only for LLM-teacher diagnostics.

Regime balancing prevents evaluation from being dominated by the most common abstract-idiosyncratic names, while color-bin balancing reduces the risk that differences are driven by over-represented hue, lightness, or saturation regions.

3.4 Interactive Color Reference Game

Each COLORREF instance is a multi-turn game between a *guesser* and a *teacher*. At turn 0, the guesser receives only the description x_i and predicts a hex code $\hat{h}_{i,0} = G(x_i)$, which is parsed as sRGB and converted to LAB, $\hat{c}_{i,0} = \text{LAB}(\hat{h}_{i,0})$. At feedback round t , the teacher observes the target c_i and current guess $\hat{c}_{i,t}$, emits $f_{i,t} = T(c_i, \hat{c}_{i,t})$, and the guesser revises its prediction as $\hat{h}_{i,t+1} = G(x_i, \hat{h}_{i,t}, f_{i,t})$. The new prediction is again converted to LAB. Games run for at most $T = 3$ feedback rounds and stop early when $\|c_i - \hat{c}_{i,t}\|_2 < 5.0$. All main experiments use temperature-0 decoding.

3.5 Metrics

The primary metric is ΔE_{76} , the Euclidean distance between target and predicted colors in CIELAB space: $e_{i,t} = \|c_i - \hat{c}_{i,t}\|_2$. Lower values indicate more perceptually similar colors, with 0 denoting an exact match. We use ΔE_{76} because it is simple, reproducible, and directly compatible with LAB coordinates; we discuss newer perceptual color-difference formulas such as CIEDE2000 as a limitation. We report initial error $e_{i,0}$, final error

$e_{i,T}$, absolute improvement $\Delta_i = e_{i,0} - e_{i,T}$, and relative improvement $\text{RI}_i = (e_{i,0} - e_{i,T}) / (e_{i,0} + \epsilon)$.

We also evaluate feedback-following behavior. **Directional alignment** measures whether the model’s update moves toward the target. At turn t , we compare the ideal correction vector $c_i - \hat{c}_{i,t}$ with the model’s actual update vector $\hat{c}_{i,t+1} - \hat{c}_{i,t}$ using cosine similarity:

$$A_{i,t} = \frac{(c_i - \hat{c}_{i,t})^\top (\hat{c}_{i,t+1} - \hat{c}_{i,t})}{\|c_i - \hat{c}_{i,t}\|_2 \|\hat{c}_{i,t+1} - \hat{c}_{i,t}\|_2}. \quad (1)$$

This value is 1 when the model moves exactly toward the target, 0 for an orthogonal update, and negative when the model moves away.

Constraint satisfaction measures whether the next guess obeys the signed constraint expressed by feedback. For example, if the teacher says *make it darker*, the next guess should reduce L^* ; if the teacher says *more blue*, it should move in the corresponding LAB direction. Unlike directional alignment, which evaluates the whole update, constraint satisfaction checks whether the model follows the specific axis-level instruction.

Finally, **behavioral geometry alignment** compares pairwise distance matrices over true LAB targets with pairwise distance matrices over model guesses or trajectories, using Spearman correlation and kernel alignment. This measures whether interaction makes model behavior more consistent with the underlying perceptual geometry.

4 Experimental Setup

We design experiments around five questions: whether feedback improves over one-shot prediction; whether bandwidth matters more than wording; whether sequential feedback helps beyond the same total information delivered at once; whether models can generate useful feedback rather than only follow it; and whether effects hold across semantic regimes and model sizes.

4.1 Teacher Conditions

The teacher determines the feedback after each prediction. We compare deterministic oracle teachers, which isolate the guesser’s ability to follow controlled perceptual corrections, with LLM teachers, which generate feedback in a more naturalistic but less controlled setting.

Oracle teachers. Oracle teachers observe the target LAB coordinate c_i and current guess $\hat{c}_{i,t}$, compute differences along LAB lightness L^* , LAB

red–green a^* , LAB yellow–blue b^* , and HSV saturation (nominally called *axis*), then rank candidate corrections by normalized absolute error after thresholding. The teacher emits up to k corrections, where k is the feedback bandwidth.

Canonical axis feedback uses short phrases such as *make it lighter*, *make it more blue*, or *make it more muted*. We use three oracle variants: a **minimal oracle** that emits exactly one canonical correction per turn, an **axis oracle** that emits one to three canonical corrections depending on bandwidth, and a **template oracle** that uses the same selected corrections but realizes them with deterministic natural-language paraphrases.

LLM teachers. We also evaluate LLM-generated feedback, instructing the teacher to use a constrained perceptual vocabulary without revealing the target hex code. We consider a **hex-only LLM teacher**, which sees the target and current guess hex codes; a **LAB-aware LLM teacher**, which also sees LAB coordinates; and an **oracle-assisted LLM teacher**, which paraphrases oracle directions into feedback. To separate teacher-generation errors from guesser-compliance errors, we include a matched-vocabulary oracle ceiling: **template oracle with LLM vocabulary**, which uses deterministic oracle directions phrased with the same vocabulary available to the LLM teacher. Prompt details appear in Appendix A.4.

4.2 Feedback Bandwidth and Information Budget

Feedback bandwidth is the maximum number of independent constraints in one teacher message, $k \in \{1, 2, 3\}$. For axis and template oracles, we run separate $k = 1$, $k = 2$, and $k = 3$ conditions. Varying k tests whether models benefit from more perceptual information per turn; comparing axis and template teachers at fixed k tests whether wording matters once bandwidth is controlled.

We also run an information-budget ablation, where conditions are matched by the total number of constraints available across the game. This lets us compare, for example, one turn with three constraints against three turns with one constraint each, testing whether iteration helps beyond total feedback quantity.

4.3 Models and Decoding

Our main experiments use Qwen/Qwen3-14B (Yang et al., 2025) as the guesser model, with

Qwen/Qwen3-8B as a smaller same-family replication. All generations use local HuggingFace Transformers (Wolf et al., 2020) inference with deterministic decoding: temperature 0, seed 13, and maximum generation length 64. The model is instructed to output a valid hex code in the form #RRGGBB. Outputs are parsed with a conservative hex extractor and converted to LAB before evaluation.

The main protocol uses at most $T = 3$ feedback rounds and convergence threshold $\Delta E < 5.0$, stopping early once the threshold is reached. A separate convergence ablation varies the maximum number of rounds to test whether performance plateaus after three turns.

4.4 Experimental Phases

Phase 1: one-shot and oracle feedback. We compare one-shot prediction from the description alone against controlled oracle feedback with up to three revision rounds.

Phase 2: bandwidth and wording. On main_1000, we isolate feedback bandwidth and wording using seven controlled oracle conditions. Here c_1 , c_2 , and c_3 indicate the maximum number of directional constraints per turn, while axis and template differ only in surface wording. The conditions are axis_{c_1} , axis_{c_2} , axis_{c_3} , template_{c_1} , template_{c_2} , template_{c_3} , and minimal_{c_1} . Examples appear in Appendix A.5.

Phase 3: information budget. We test whether iteration matters at fixed feedback budget by comparing matched-budget conditions, such as one turn with three constraints versus three turns with one constraint each.

Phase 4: LLM-teacher decomposition. On debug_4000, we run the LLM-teacher variants and the matched-vocabulary oracle ceiling, then audit generated feedback by detecting directional constraints and checking them against the true target–guess geometry.

Phase 5: diagnostic ablations and replication. We run three diagnostic studies: a longer-turn convergence sweep, an in-context calibration and LoRA finetuning analysis, and a model-size comparison between Qwen3-14B and Qwen3-8B on selected conditions.

Condition	N	Init ΔE	Final ΔE	Rel. imp.
one-shot	12,000	40.58	40.58	0.000
axis _{c3}	12,000	40.58	22.32	0.262
template _{c3}	12,000	40.58	22.88	0.243
minimal _{c1}	12,000	40.58	28.29	0.124
template-vocab	12,000	40.58	29.68	0.120

Table 3: Main 12k results for Qwen3-14B. Controlled feedback substantially reduces LAB error relative to one-shot prediction.

4.5 Reported Quantities

For all conditions, we report mean initial error, mean final error, relative improvement, convergence rate, directional alignment, and constraint satisfaction where applicable. For the main 12,000-example results and regime analyses, we report percentile bootstrap 95% confidence intervals using 1,000 resamples with seed 13. Diagnostic analyses additionally report task-specific quantities such as best-so-far error, total constraint budget, ICL demonstration count, and LLM-teacher directional precision.

5 Results

We evaluate COLORREF along six dimensions: improvement over one-shot prediction, feedback bandwidth and temporal structure, semantic-regime variation, LLM-teacher feedback generation, behavioral geometry, and diagnostic ablations.

5.1 Controlled feedback improves color grounding

We first compare one-shot prediction against controlled feedback on the 12,000-example scale-up set. The one-shot baseline predicts a hex code from the description alone; feedback conditions allow up to three revision rounds.

Table 3 shows that controlled feedback substantially improves color grounding. One-shot prediction yields mean LAB error 40.58. With three-constraint axis feedback, final error drops to 22.32, a relative improvement of 26.2%; template feedback with the same bandwidth reaches 22.88. Single-constraint feedback also improves over one-shot but is weaker, ending around 28–30 ΔE .

The effect is stable at scale: the 95% bootstrap interval for axis_{c3} final error is [21.85, 22.76], and for template_{c3} it is [22.38, 23.36]. Qualitatively, trajectories show that even when the initial association is wrong, local perceptual feedback can steer the model toward the intended color.

Condition	Max c	Final ΔE	Rel. imp.	C-sat	Align.
axis _{c1}	1	28.13	0.127	0.935	0.549
axis _{c2}	2	22.78	0.266	0.936	0.727
axis _{c3}	3	21.94	0.254	0.898	0.742
template _{c1}	1	29.97	0.128	0.889	0.483
template _{c2}	2	23.40	0.255	0.922	0.703
template _{c3}	3	22.32	0.266	0.876	0.712
minimal _{c1}	1	28.13	0.127	0.935	0.549

Table 4: Bandwidth ablation on main_4000. Increasing feedback from one to two constraints gives the largest gain; axis and template wording are similar at matched bandwidth.

Condition	Rounds	c /turn	Budget	Final ΔE
one-turn _{c2}	1	2	2	31.98
two-turn _{c1}	2	1	2	29.71
one-turn _{c3}	1	3	3	31.88
three-turn _{c1}	3	1	3	28.13
three-turn _{c3}	3	3	9	21.94

Table 5: Information-budget ablation on main_4000. At matched total constraint budgets, iterative feedback outperforms delivering the same number of constraints in a single turn.

5.2 Bandwidth matters, but iteration matters too

We next isolate feedback *bandwidth*, the number of independent constraints per turn, and *temporal structure*, whether constraints are delivered at once or sequentially.

Table 4 shows that bandwidth is the dominant factor among oracle-feedback variants. Moving from one to two constraints reduces final error by roughly 5–7 ΔE : axis feedback improves from 28.13 to 22.78, while template feedback improves from 29.97 to 23.40. Moving from two to three constraints yields a smaller additional gain. Wording has a weaker effect once bandwidth is controlled: at $c = 3$, axis and template feedback are close, 21.94 versus 22.32 final ΔE .

Table 5 shows that temporal structure also matters. With the same total budget of three constraints, three one-constraint turns reach 28.13 final error, while one three-constraint turn reaches only 31.88. The same pattern holds at budget two: two one-constraint turns reach 29.71, compared with 31.98 for one two-constraint turn. Thus, feedback is not merely an information payload: later corrections are useful because they are conditioned on the model’s revised guess.

Regime	N	Init ΔE	Final ΔE	Rel. imp.
abstract	3,000	49.92	24.05	0.343
compound	3,000	36.87	21.96	0.237
explicit	3,000	36.02	21.15	0.246
prototype	3,000	39.51	22.14	0.222

Table 6: Regime effects for axis_{c3} on main_12000. Abstract descriptions have the highest initial error but the largest relative improvement.

5.3 Semantic regimes reveal difficulty and recoverability

The semantic-regime taxonomy lets us ask whether different kinds of color descriptions benefit differently from interaction. Table 6 reports axis_{c3} results on main_12000.

Abstract-idiosyncratic descriptions are hardest initially but most recoverable under feedback. Their one-shot error is 49.92, far above the other regimes, but final error drops to 24.05, yielding the largest relative improvement, 34.3%. Explicit-grounded names start closer to the target and end with the lowest final error, 21.15, while prototype-mediated and compound-associative descriptions fall between these cases.

This complicates a simple “abstract names are noise” interpretation. Abstract descriptions are weakly grounded in one shot, but many become recoverable with informative perceptual feedback. The full regime-by-teacher table shows the same pattern: axis_{c3} and template_{c3} reduce abstract-name error by roughly half, whereas single-constraint teachers help less.

5.4 Following feedback is easier than generating feedback

Oracle experiments isolate the guesser’s ability to follow correct feedback. We next ask whether LLMs can generate such feedback themselves. Table 7 compares LLM-teacher variants on debug_4000.

The matched-vocabulary oracle ceiling performs best, with final error 28.84 and constraint satisfaction 0.886. Because it uses deterministic oracle directions phrased with the vocabulary available to LLM teachers, its success shows that the guesser can use this feedback style when the underlying direction is correct.

LLM teachers are substantially weaker. The LAB-aware teacher performs best among generated-feedback variants, reaching final error 31.64 and positive relative improvement. The

Teacher	ΔE	Rel. imp.	C-sat	DP
template-vocab oracle	28.84	0.118	0.886	–
LLM lab-aware	31.64	0.091	0.456	0.39
LLM hex-only	35.56	-0.049	0.487	0.31
LLM oracle-assisted	35.66	-0.083	0.651	0.11

Table 7: LLM-teacher decomposition on debug_4000. The matched-vocabulary oracle ceiling shows that the guesser can follow well-formed feedback; LLM teachers often fail to generate geometrically correct feedback. Here, ΔE denotes final ΔE , and DP denotes directional precision.

Condition	ρ init	ρ final	ρ traj.	Kernel final
one-shot	0.461	0.461	0.461	0.508
axis _{c3}	0.461	0.796	0.804	0.832
template _{c3}	0.461	0.764	0.789	0.805
minimal _{c1}	0.461	0.693	0.696	0.717
template-vocab	0.461	0.663	0.666	0.681

Table 8: Behavioral geometry alignment on main_12000. Feedback makes the geometry of model guesses and trajectories more aligned with true LAB geometry.

hex-only teacher performs worse, suggesting that hex values alone are insufficient for reliable feedback generation. The oracle-assisted LLM teacher also performs poorly: despite higher constraint satisfaction, its audited directional precision is only 0.11, indicating that paraphrased feedback often expresses the wrong geometric direction.

These results decompose interactive grounding into two capabilities. The guesser can often follow clear perceptual feedback, but generating feedback requires translating a target–guess difference into geometrically valid language. The main LLM-teacher failure mode is therefore not leakage or formatting, but fluent, misdirected feedback.

5.5 Feedback improves behavioral geometry alignment

Beyond pointwise error, we ask whether feedback improves the global geometry of model behavior. For each condition, we compare pairwise distance matrices over true LAB targets with pairwise distance matrices over model guesses or trajectories, using Spearman correlation and kernel alignment.

Table 8 shows that one-shot guesses have moderate alignment with true LAB geometry, with Spearman $\rho = 0.461$. Feedback substantially increases this alignment: axis_{c3} reaches final-guess alignment $\rho = 0.796$ and trajectory-state alignment

$\rho = 0.804$, while template_{c3} reaches $\rho = 0.764$ and $\rho = 0.789$. Single-constraint feedback also improves alignment, but less strongly. Thus, feedback not only reduces individual errors; it makes pairwise relationships among model outputs more faithful to perceptual color space.

5.6 Additional ablations: convergence, ICL, finetuning, and model size

Additional ablations clarify the scope of the main results. First, a longer-turn sweep shows that final error improves rapidly from one to three turns and then mostly plateaus: five and ten turns reach 21.43 and 21.04 final ΔE , while the final–best gap grows, indicating overshooting (Appendix B.1). Second, in-context examples modestly improve one-shot prediction, especially for abstract-idiosyncratic names, but add little once oracle feedback is available: feedback variants remain around 21.8–22.3 final ΔE (Appendix B.2). Parameter-efficient LoRA finetuning on a balanced 4k subset likewise yields no substantial change, suggesting that per-example perceptual feedback is a stronger localization signal than small-scale calibration. Finally, Qwen3-8B shows the same qualitative ordering as Qwen3-14B but benefits less from feedback, so we treat the model-size comparison as a same-family replication rather than a scaling claim (Appendix B.4).

6 Discussion

Interaction as adaptive grounding. The information-budget ablation shows that feedback is not merely an additional information channel. At matched total constraint budgets, sequential feedback outperforms delivering the same constraints in a single turn because later teacher messages are conditioned on the model’s previous guess. In this sense, COLORREF tests whether language can act as a control signal for navigating perceptual space: a correction such as *more blue* is useful only relative to the current guess, and changes as the guess changes.

Why three turns? The longer-turn sweep supports three feedback rounds as the main protocol: final error improves rapidly from one to three turns and then largely plateaus. Longer games reduce best-so-far error, but the growing final–best gap indicates overshooting. Three turns therefore provide a practical compute–quality tradeoff, while longer games are better suited for stopping policies or best-state selection.

Demonstrations calibrate priors, feedback localizes targets. In-context examples modestly improve one-shot predictions, especially for abstract-idiosyncratic descriptions, but add little once oracle feedback is available. This suggests that demonstrations calibrate the initial color prior, whereas per-example feedback supplies the stronger localization signal. The LoRA result points in the same direction: small-scale supervision does not substantially change the conclusions, indicating that the main gains come from adaptive feedback rather than dataset calibration alone.

Following and generating feedback are distinct capabilities. The LLM-teacher results reveal a sharp asymmetry: the guesser can often follow well-formed perceptual corrections, but LLM teachers frequently generate fluent feedback that points in the wrong geometric direction. This is a controlled form of common-ground misalignment, where the teacher produces a plausible utterance that is not aligned with the perceptual relation needed for update (Sarkar et al., 2025). This matters for agentic and self-refinement settings: a model that benefits from good feedback may still be a poor teacher or critic. COLORREF makes this failure mode measurable through directional precision, constraint satisfaction, and paired trajectories.

Behavioral geometry as a complement to representation probing. Prior work asks whether internal representations align with perceptual color geometry. COLORREF asks whether outputs and trajectories can be steered toward that geometry. The increase in pairwise geometry alignment after feedback suggests that interaction reshapes not only individual predictions but also the global structure of model behavior, linking grounded semantic evaluation with representation-geometry analysis.

7 Conclusion and Future Work

We introduced COLORREF, an interactive reference-game framework for probing language-to-color grounding beyond one-shot prediction. The results show that controlled feedback improves grounding, bandwidth and sequential adaptation matter, abstract descriptions benefit strongly from interaction, and models can follow valid perceptual feedback better than they can generate it. Future work should extend this approach to multilingual data, human participants, distributional color targets, and other continuous perceptual domains.

8 Limitations

Our study has several limitations. First, all main experiments use English color descriptions. Extending COLORREF to multilingual data will require distributional comparisons rather than parallel item-level matching, since color naming systems and naturally occurring descriptions differ across languages. Second, the scaled experiments use Qwen3-14B, with Qwen3-8B as a same-family replication; broader claims require evaluation across additional model families and multi-modal systems. Third, the LLM-teacher experiments are diagnostic-scale, using debug_4000, so they identify a failure mode rather than exhaustively benchmarking teacher models. Fourth, oracle teachers are controlled probes that provide geometrically valid feedback by construction; they should not be interpreted as models of human tutoring. Finally, we evaluate against single crowd-sourced sRGB targets using Euclidean distance in CIELAB (ΔE_{76}). Abstract descriptions may evoke distributions of plausible colors across speakers, and future work should incorporate human judgments and alternative perceptual metrics such as CIEDE2000.

References

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A Prompt Templates

This section reports the prompt templates used for the guesser and teacher conditions. All generations use deterministic decoding with temperature 0.

A.1 One-shot and Initial Guess Prompt

The one-shot baseline and the first turn of feedback games use the same template.

```
You are playing a color guessing game.

Color description:
"{raw_name}"

Guess the intended color as a hex code.
Answer only with a valid hex code in the form
#RRGGBB.
```

A.2 Revision Prompts

After each teacher message, the guesser receives the original description, its previous guess, and the

latest feedback. The default prompt uses only the most recent revision step:

```
You are playing a color guessing game.

Color description:
"{raw_name}"

Your previous guess was: {previous_guess_hex}
Feedback: {feedback}

Revise your guess.
Answer only with a valid hex code in the form
#RRGGBB.
```

We also evaluate a history-aware revision prompt, where the guesser receives the full interaction history for the current game rather than only the latest feedback message:

```
You are playing a color guessing game.

Color description:
"{raw_name}"

Game history:
Turn 0 guess: {guess_0_hex}
Feedback 0: {feedback_0}

Turn 1 guess: {guess_1_hex}
Feedback 1: {feedback_1}

Turn 2 guess: {guess_2_hex}
Feedback 2: {feedback_2}

Revise your guess using the full history above.
Answer only with a valid hex code in the form
#RRGGBB.
```

A.3 Oracle Feedback Templates

Oracle teachers compute target–guess differences in LAB lightness L^* , LAB red–green a^* , LAB yellow–blue b^* , and HSV saturation. Candidate axes are filtered by minimum thresholds and ranked by normalized absolute error. The axis oracle emits up to k corrections per turn, where $k \in \{1, 2, 3\}$.

Example oracle feedback messages are:

- one constraint: *Make it lighter.*
- two constraints: *Make it lighter and more blue.*
- three constraints: *Make it lighter, more blue, and more muted.*

The minimal oracle is the special case $k = 1$. The template oracle uses the same selected axes as the axis oracle, but realizes each direction with a deterministic paraphrase chosen from a small template set.

A.4 LLM Teacher Prompts

The LLM teacher receives the target color and the current guess, and is asked to provide concise feedback without revealing the target. The prompt uses a constrained perceptual vocabulary including *lighter/darker, more red/more green, more yellow/more blue, more saturated/more muted, and warmer/cooler*. In the LAB-aware condition, the teacher additionally receives target and guess LAB coordinates. In the oracle-assisted condition, oracle-selected directions are provided and the LLM is asked to paraphrase them.

Main LLM teacher used:

You are the teacher in a color reference game.

Target color description:
"{raw_name}"

True target color: {target_hex}
Current guess: {guess_hex}

Give one short piece of comparative feedback to help the guesser move closer to the true target.

Use only these feedback types:

- lighter or darker
- more red or more green
- more yellow or more blue
- more saturated or more muted
- warmer or cooler

Use at most {max_clauses} feedback clause(s) in one short sentence.

Do not reveal the target hex code.
Do not mention a new object example.
Do not say the exact answer.
Answer with one short sentence only.

The Oracle-assisted variant:

You are the teacher in a color reference game.

Target color description:
"{raw_name}"

True target color: {target_hex}
Current guess: {guess_hex}

The correct perceptual corrections for this step are (paraphrase only, do not list as JSON): {oracle_constraints_json}

Turn these into one natural short sentence of comparative feedback for the guesser.

Use only these feedback types:

- lighter or darker
- more red or more green
- more yellow or more blue
- more saturated or more muted
- warmer or cooler
- closer to gray or less gray

Use at most {max_clauses} feedback clause(s).

Do not reveal the target hex code.
Do not mention exact RGB or LAB values.
Do not mention a new object example.
Do not say the exact answer.
Answer with one short sentence only.

The LAB variant:

You are the teacher in a color reference game.

Target color description:
"{raw_name}"

True target color: {target_hex}
Target LAB (L, a, b): {target_lab_l}, {target_lab_a}, {target_lab_b}

Current guess: {guess_hex}
Current guess LAB (L, a, b): {guess_lab_l}, {guess_lab_a}, {guess_lab_b}

Give one short piece of comparative feedback to help the guesser move closer to the true target.

Use only these feedback types:

- lighter or darker
- more red or more green
- more yellow or more blue
- more saturated or more muted
- warmer or cooler
- closer to gray or less gray

Use at most {max_clauses} feedback clause(s) in one short sentence.

Do not reveal the target hex code.
Do not quote exact LAB numbers in your answer.
Do not mention a new object example.
Do not say the exact answer.
Answer with one short sentence only.

A.5 Bandwidth and Wording Conditions

The controlled oracle conditions vary two factors: feedback *bandwidth*, the maximum number of constraints emitted in one turn, and feedback *wording*, the surface form used to express those constraints. The subscript c_1 , c_2 , or c_3 denotes the maximum number of constraints per teacher message.

For example, suppose the target is lighter, bluer, and more muted than the current guess. The conditions differ as follows:

Condition	Example feedback
minimal _{c1}	<i>Make it lighter.</i>
axis _{c1}	<i>Make it lighter.</i>
axis _{c2}	<i>Make it lighter and more blue.</i>
axis _{c3}	<i>Make it lighter, more blue, and more muted.</i>
template _{c1}	<i>Try making it a little lighter.</i>
template _{c2}	<i>It should be lighter and closer to blue.</i>
template _{c3}	<i>Move it lighter, bluer, and a bit more muted.</i>

The **axis** conditions use canonical direction phrases with fixed wording, while the **template**

conditions use the same selected directions but realize them with deterministic natural-language paraphrases. The **minimal** condition is a one-constraint oracle used as a simple baseline; in practice it is equivalent to the one-constraint axis oracle in our controlled setting, but we keep the name to distinguish it from the bandwidth sweep.

B Additional Results

B.1 Longer-Turn Convergence

The main protocol uses up to three feedback rounds. To test whether performance continues improving with longer games, we run an axis_{c3} turn sweep on main_{4000} . Table 9 reports final error, best-so-far error, and regression rate.

Turns	Final ΔE	Best ΔE	Final-best	Reg. rate
1	31.88	26.48	5.40	28.5%
2	24.89	19.46	5.43	22.3%
3	21.94	15.67	6.27	19.6%
5	21.43	12.39	9.04	20.2%
7	20.62	11.16	9.46	17.4%
10	21.04	10.17	10.87	15.7%

Table 9: Longer-turn convergence sweep for axis_{c3} on main_{4000} . Final error mostly plateaus after three turns, while best-so-far error continues to improve but is increasingly not retained.

The largest gains occur early. Best-so-far error improves by 15.00 ΔE from turn 0 to 1, 7.02 from turn 1 to 2, and 3.79 from turn 2 to 3. After turn 3, marginal gains fall quickly: 2.03 from turn 3 to 4, 1.25 from turn 4 to 5, and below 1.0 thereafter. This supports using three turns as a compute-quality tradeoff for final-answer evaluation.

B.2 In-Context Calibration

We test whether providing examples of color-description/hex pairs improves the model’s color prior. Table 10 reports one-shot and feedback-game results by number of demonstrations.

k	One-shot ΔE	Feedback init ΔE	Feedback final ΔE
0	41.47	41.47	21.94
2	38.53	38.53	21.77
4	38.71	38.71	22.00
8	39.25	39.25	22.29

Table 10: In-context calibration on main_{1000} . Demonstrations modestly improve one-shot prediction, but provide little additional benefit once oracle feedback is available.

Averaged over $k = 2, 4, 8$, random and regime-matched examples yield similar one-shot errors, 38.23 and 38.27 respectively, while text-similar examples are worse at 40.00. This suggests that lexical similarity between color names does not necessarily imply perceptual similarity. The largest one-shot ICL gain occurs for abstract-idiosyncratic descriptions, where mean error drops from 54.21 to 47.92.

B.3 Information-Budget Ablation

Table 11 reports the full information-budget ablation on main_{12000} . Conditions vary the number of feedback rounds, maximum constraints per turn, and total constraint budget.

Condition	Rounds	c/turn	Budget	Final ΔE
one-turn $_{c1}$	1	1	1	33.81
one-turn $_{c2}$	1	2	2	31.98
one-turn $_{c3}$	1	3	3	31.88
two-turn $_{c1}$	2	1	2	29.71
two-turn $_{c2}$	2	2	4	25.11
three-turn $_{c1}$	3	1	3	28.13
three-turn $_{c3}$	3	3	9	21.94

Table 11: Information-budget ablation. At matched total budgets, iterative feedback outperforms delivering the same number of constraints in a single turn.

At budget 3, three one-constraint turns reach final error 28.13, while one three-constraint turn reaches 31.88. At budget 2, two one-constraint turns reach 29.71, while one two-constraint turn reaches 31.98. Thus, interaction adds value beyond total feedback quantity.

B.4 Model-Size Comparison

Table 12 reports a same-family model-size comparison on main_{4000} .

Model	Condition	Final ΔE	Rel. imp.
Qwen3-14B	one-shot	41.47	0.000
Qwen3-14B	axis_{c3}	21.94	0.254
Qwen3-14B	template $_{c1}$	29.97	0.128
Qwen3-8B	one-shot	44.18	0.000
Qwen3-8B	axis_{c3}	27.99	0.092
Qwen3-8B	template $_{c1}$	35.54	-0.038

Table 12: Model-size comparison. Qwen3-8B shows the same qualitative ordering as Qwen3-14B, but benefits less from feedback.

C Dataset Construction Details

C.1 Preprocessing Pipeline

We construct COLORREF from a ColorNames-derived corpus of user-provided color names paired with sRGB hex codes. The pipeline proceeds in four stages.

1. **Dataset filtering.** We load source color-name entries and LLM-derived validity scores. The validity score estimates whether a string functions plausibly as a color description. We retain English entries with score greater than 0.75 and valid hex codes. The data was analyzed to detect any PII information.
2. **Canonicalization.** We assign stable example identifiers, normalize basic string fields, tokenize descriptions, and compute length features.
3. **Color conversion.** We convert target hex codes to RGB, HSV, and CIELAB coordinates. Main evaluation uses Euclidean distance in CIELAB space.
4. **Taxonomy construction.** We run a lightweight parser and compute component scores for explicitness, prototype anchoring, rarity, and abstraction. These scores are used to assign each description to a semantic regime.

The final corpus contains 518,830 English examples. Each row contains the raw color description, target hex code, RGB/HSV/CIELAB coordinates, parser outputs, component scores, regime label, and perceptual bins.

C.2 Color-name structural parser

We assign each raw name a structural parse using a deterministic, lexicon-driven procedure (no model weights). Strings are normalized to lowercase; apostrophes, hyphens, and quotes are mapped to spaces; non-alphabetic characters are removed; runs of whitespace are collapsed. The normalized string is split on spaces into tokens. Head selection scans tokens from right to left: the first token in a fixed head lexicon of basic color terms is taken as the head. Tokens strictly before the head are partitioned into known modifiers (from a fixed modifier lexicon of intensity, temperature, texture, and similar non-head descriptors) versus unknown prefix tokens that are neither heads nor listed modifiers.

From this decomposition we assign one of several parse categories: for example, a single head token only, exactly one known modifier with no unknown prefixes, multiple known modifiers and no unknown prefixes, a head with at least one unknown prefix, or no recognized head. We also assign a parse confidence that is high when the prefix is fully explained by the modifier lexicon and decreases when unknown prefixes appear (capped with a floor for compound cases).

Parser v2 (used downstream). For experiments that require broader coverage, we use an extended parser that applies token-level rules before lexicon lookup: *-ish* normalization (e.g., mapping greenish to green, with a small exception table), optional plural stripping when the singular stem is a known head or prototype, and—if no basic color head is found—a second right-to-left pass that allows prototype heads (object-, material-, or nature-associated color words from a separate lexicon). That yields additional categories such as prototype-only heads, modifiers with a prototype head, and compounds around a prototype head. Regardless of whether the final head is basic or prototypical, v2 also records all prototype lexicon hits anywhere in the name as auxiliary candidates for later taxonomy features. All lexicons are hand-curated YAML lists checked into the repository.

C.3 Validity Filtering

The validity filter removes entries judged unlikely to function as color descriptions. We used GPT 4o as a judge with the following prompt (extract):

You are evaluating whether text entries are plausible color descriptions. You will receive a list of descriptions and must score each one from 0 to 1, where:

- ****1.0**** = Definitely a plausible color description
- ****0.5**** = Uncertain / borderline case
- ****0.0**** = Not a meaningful color description

What to ACCEPT (score more than 0.7)

****Literal color descriptions****: "dark red", "light blue", "pale yellow"
****Figurative/associative descriptions****: "sunset", "ocean", "grandmother's kitchen", "tired Monday morning"
****Subjective/abstract descriptions****: "melancholy", "peaceful", "nostalgic", "hopeful"

What to REJECT (score less 0.3)

```

1129    **Spam/promotional**: "buy now",
1130    "visit my site", "subscribe", "click here"
1131    **URLs/emails**: "www.example.com",
1132    "user@email.com", any web links
1133    **Nonsensical text**: "asdfgh",
1134    "qwerty", "zzzzz", random characters
1135    **Non-descriptive statements**: "im a genius",
1136    "hello world", "test 123"
1137    **Personal messages**: "love you mom",
1138    "happy birthday", "call me"
1139
1140    ## Borderline Cases (score 0.4-0.6)
1141
1142    **Ambiguous single words**: Could be a
1143    name or a description? Use context.
1144    **Very abstract concepts**: "philosophy",
1145    "mathematics" (unless clearly sensory)
1146    **Proper names**: "Jennifer", "Tokyo"
1147    (unless commonly used for colors)
1148    **Mixed content**: Partially meaningful
1149    but with errors/spam elements
1150
1151

```

The threshold is fixed at score > 0.75 . This filter is used to remove obvious noise while preserving creative and abstract color names. The score is not used as a target variable in the experiments.

D AI Assistance Disclosure

We used AI coding assistance, specifically Claude Code, to automate routine experiment-management tasks, including launching ablation runs, organizing outputs, and checking generated result files. All experimental designs, metrics, analyses, and paper claims were specified and reviewed by the authors.